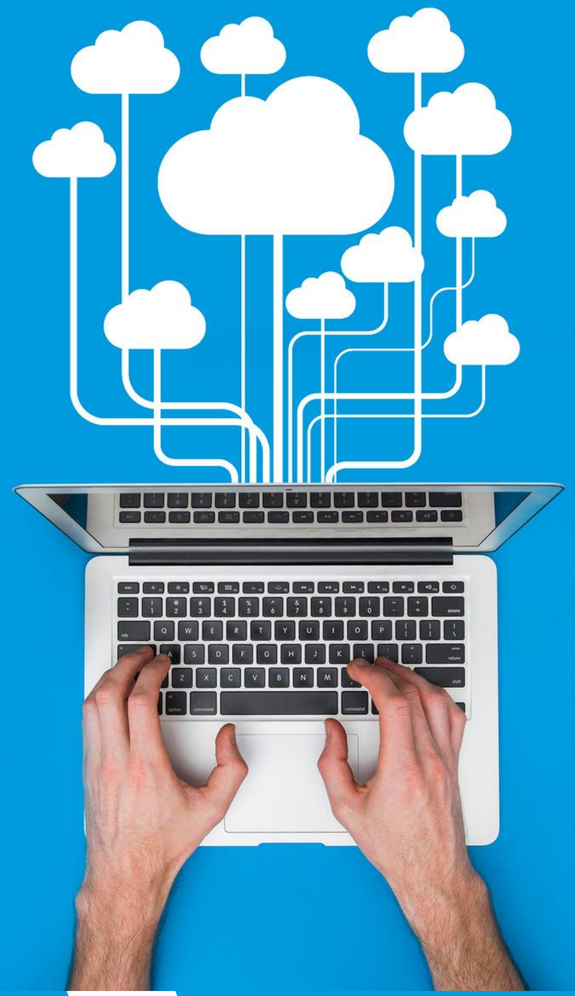
PYTHON PROGRAMMING



21 Days Challenge

Day-05



Day-05

LISTS - ACCESS, UPDATE, ITERATE, COMPREHEND, AND MANIPULATE DATA

Instructor



Abdullah Al Zubaer
Instructor – Artificial Intelligence & Data Science
B.Sc. in Internet of Things and Robotics Engineering
Gazipur Digital University, Gazipur, Bangladesh

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List

A list is a built-in data type in Python that is used to store multiple items in a single variable.

```
FRUITS = ['ORANGE','PEAR', 'BANANA','APPLE', 'BANANA']
```

- ORDERED
- CHANGEABLE (MUTABLE)
- ALLOWS DUPLICATES

Accessing Lists Item

```
list = ['physics', 'chemistry', 1997, 2000];
print(list[0]) #indexing
print(list[-4]) # negative indexing
print(list[:-3]) #slicing
```

Output

```
physics
physics
['physics']
```

Update List Items

```
list = [1, 2, 3, 4, 5]
list[3]=3
print(list)

list1 = ['x','y','z']
list[1:3]=list1
print(list)
```

Output

```
[1, 2, 3, 3, 5]
[1, 'x', 'y', 'z', 3, 5]
```

Add List Items

```
my_list = [1, 2, 3]

my_list.append(4)
print(my_list)

my_list.insert(1, 'a')
print(my_list)

my_list.extend([4, 5])
print(my_list)
```

Output

```
[1, 2, 3, 4]
[1, 'a', 2, 3, 4]
[1, 'a', 2, 3, 4, 4, 5]
```

Remove List Items

```
fruits = ['apple', 'banana', 'apple', 'orange']

fruits.remove('apple')
print(fruits)

fruits.pop(1)
print(fruits)

fruits.clear()
print(fruits)
```

Output

```
['banana', 'apple', 'orange']
['banana', 'orange']
[]
```

Remove List Items

```
fruits = ['apple', 'banana', 'apple', 'orange']

del fruits[1]
print(fruits)

del fruits[1:3]
print(fruits)

del fruits
#print(fruits)
```

Output

```
['apple', 'apple', 'orange']
['apple']
```

List Comprehension

A short and clean way to create a list from an existing iterable.

```
list = [expression for item in iterable if condition]
```

```
squares = [x*x for x in range(5)]
print(squares)
```

Output

```
[0, 1, 4, 9, 16]
```

Iterate Using List Comprehension

```
fruits = ['apple', 'banana']

upper_fruits = [fruit.upper() for fruit in fruits]
print(upper_fruits)
```

Output

```
['APPLE', 'BANANA']
```

QUESTION SESSION

THANK YOU

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